

Step 1: State the hypotheses

Define the claim in terms of the population mean (μ). Use the null hypothesis as the default position.

- $H_0: \mu = 3$ hours (the population mean daily TV watching time is 3 hours).
- $H_1: \mu \neq 3$ hours (the population mean daily TV watching time differs from 3 hours).
- Test type: two-tailed (the difference could be above or below 3 hours).

Step 2: Set the decision criteria

Choose the significance level (α) before you evaluate the sample result.

- $\alpha = 0.05$.
- Critical values for a two-tailed Z-test at $\alpha = 0.05$: $Z = \pm 1.96$.
- Decision rule: reject H_0 if $Z < -1.96$ or $Z > +1.96$; otherwise, do not reject H_0 .

Step 3: Compute the test statistic

Use the Z-test for a mean when the population standard deviation (σ) is treated as known, as shown in the notes.

Test statistic:

- $Z = (\bar{x} - \mu_0) / (\sigma / \sqrt{n})$

Where:

- $\bar{x}$ is the sample mean.
- μ_0 is the hypothesised mean under H_0 (here, 3 hours).
- σ is the population standard deviation.
- n is the sample size.

In the provided completed example, the computed Z value is 4.0 (based on a sample mean of 4 hours versus $\mu_0 = 3$ hours).

Step 4: Make the decision

Compare the observed Z to the critical values:

- Observed $Z = 4.0$.
- Critical values = ± 1.96 .

Because 4.0 is greater than 1.96, the result lies in the rejection region. The associated two-tailed p-value is far below 0.05, so the result is statistically significant.

Step 5: Write the conclusion

State the decision and interpret it in context using cautious statistical language.

- Decision: Reject H_0 at $\alpha = 0.05$.
- Conclusion: The sample provides strong evidence that the population mean daily TV watching time is not 3 hours.
- Direction (based on the example sample mean): the evidence points to a mean above 3 hours.

Optional short note (if your worksheet asks about errors)

A hypothesis test can make two main types of mistakes:

- Type I error (α): rejecting a true H_0 (concluding $\mu \neq 3$ when μ is actually 3).
- Type II error (β): not rejecting a false H_0 (missing a real difference from 3).